Ankit Deshmukh

Ph.D. • Water Resources

deshmukh.acad@gmail.com • 096305 21359 • www.ankitdeshmukh.com • anixn

Summary

Assistant Professor at Pandit Deendayal Energy University (PDEU) with a Ph.D. in Water Resources from IIT Hyderabad (2021). Research interests include computational hydrology, drought modeling, catchment classification, remote sensing, GIS, and machine learning for water resources. Current work focuses on developing computational frameworks for drought forecasting and water resource vulnerability assessment under climate variability, including contributions to an ICSSR-funded project on flood and drought early warning systems. Proficient in R, Python, and Google Earth Engine.

Education

Ph.D. **Indian Institute of Technology Hyderabad**, Ph.D. in Water Resource – Hyderabad, India — Jan 2016 – Jan 2021
- **Topic:** Assessing vulnerability of catchments to environmental changes
- **Supervisors:** Dr. Riddhi Singh

M.Tech. **Indian Institute of Technology Hyderabad**, M.Tech. in Water Resources Engineering – Hyderabad, India — July 2014 – Jan 2016
- **Topic:** Physio-climatic controls on vulnerability of watersheds to climate and land use change across the United States
- **Supervisors:** Dr. Riddhi Singh
- **CGPA:** 8.87

B.E. **Bansal Institute of Science and Technology, Rajiv Gandhi Proudyogiki Vishwavidyalaya**, B.E. in Civil Engineering – Bhopal, India — July 2009 – Aug 2013
- **Topic:** Study of Rainwater Harvesting System at BIST Bhopal
- **Supervisors:** Prof. Maroof Khan
- **Percentage:** 76.6% with distinction

Experience

Pandit Deendayal Petroleum University, Assistant Professor – Gujarat, India — Jan 2021 – present
Department of Civil Engineering
- Teaching: Hydrology & Water Resources, Advanced Surveying & Geomatics, Remote sensing and Geographical Information System, Geospatial Analysis, Basic Surveying

Research Interests

- Hydrologic modeling, Surface water hydrology & Computational hydrology
- Catchment classification, Climate change modeling
- Vulnerability framework, Physio-climatic database
- Remote Sensing and GIS

Publications

Stochastic modeling-based queueing frameworks for performance enhancement and economic optimization of hybrid electric vehicle charging networks — 2026
Shreekant Varshney, Bhasuru Abhinaya Srinivas, Sherwin Abraham, Manthan Shah, *Ankit Deshmukh*, Kaibalya Prasad Panda, Mohit Bajaj, Vojtech Blazek, Lukas Prokop
doi.org/10.1016/j.rineng.2026.111066 (Results in Engineering, Volume 30, 111066, ISSN 2590-1230)

Novel control strategies for electric vehicle charging stations using stochastic modeling and queueing analysis — 2025
Shreekant Varshney, Kaibalya Prasad Panda, Manthan Shah, Bhasuru Abhinaya Srinivas, *Ankit Deshmukh*, Kapil Kumar Choudhary, Mohit Bajaj, Lukas Prokop, Olena Rubanenko
10.1038/s41598-025-04725-7 (Scientific Reports 15, no. 1 (2025): 21391)

Prediction of Hydrological Drought Pattern and Duration in Data Scarce Catchments with Catchments' Physio-Climatic Attributes Dec 2023
Ankit Deshmukh
(International Conference on Hydraulics, Water Resources and Coastal Engineering (Springer Nature Singapore))

Sustainable Modeling Framework for the Prediction in Ungauged Catchments: Leveraging Large-Sample Hydrology Datasets with Machine Learning Feb 2025
Ankit Deshmukh, S. Thanki, D. Sharma
(2025 International Conference on Sustainable Energy Technologies and Computational Intelligence (SETCOM), IEEE)

A Sustainable Framework for Drought Modeling using Meteorological Drought Indicators as Proxies for Hydrological Drought Feb 2025
Ankit Deshmukh, G. Najwani, S. Kumari, C. Tirupathi
(2025 International Conference on Sustainable Energy Technologies and Computational Intelligence (SETCOM), IEEE)

An Enhanced Bottom-Up Approach to Assess the Catchments' Vulnerability to Climate Change Aug 2022
V. Rakhecha, *Ankit Deshmukh*
(International Conference Innovation in Smart and Sustainable Infrastructure (Springer Nature Singapore))

Physio-climatic controls on vulnerability of watersheds to climate and land use change across the United States 2016
Ankit Deshmukh, R. Singh
(Water Resources Research, doi: 10.1002/2016WR019189)

A Whittaker-Biome based framework to account for the impact of climate change on catchment behavior 2019
Ankit Deshmukh, R. Singh
(Water Resources Research, doi: 10.1029/2018WR023113)

Comparison of image processing techniques to identify the land use/ land cover changes in the Indian semi-arid region Aug 2022
S. Kumari, *Ankit Deshmukh*
(Innovation in Smart and Sustainable Infrastructure conference, Pandit Deendayal Energy University, Gandhinagar)

Impact of Climate Change on Meteorological Drought in Different Climate Zones Across the Indian Sub-Continent Dec 2020
S. Kumari, *Ankit Deshmukh*
(Fall Meeting of the American Geophysical Union, San Francisco, USA)

Discovering linkages between catchment characteristics and water quality using catchment classification Sept 2019
Ankit Deshmukh, R. Singh
(Water Future International Conference, Bengaluru, India)

Catchment classification: a tool to understand hydrology in data scarce regions June 2018
R. Singh, *Ankit Deshmukh*, A. Samal
(15th Annual Meeting of the Asia Oceania Geosciences Society (AOGS), Honolulu, Hawaii, USA)

Identifying physio-climatic controls on watershed vulnerability to climate and land use change Jan 2018
Ankit Deshmukh, R. Singh
(International Soil and Water Assessment Tool Conference, Chennai, India)

Towards a robust framework for catchment classification Dec 2017
Ankit Deshmukh, S. Samal, R. Singh
(Fall Meeting of the American Geophysical Union, New Orleans, USA)

How do watersheds physio-climatic characteristics control its vulnerability to climate and land use change? 2016
Ankit Deshmukh, R. Singh, Veena Sai
(NWMCCAAAS, MANAGE, Hyderabad, India)

What controls vulnerability of watersheds to climate and land use change across the United States? Dec 2015
R. Singh, *Ankit Deshmukh*
(Fall Meeting of the American Geophysical Union, San Francisco, USA)

Book Chapters

Assessing the vulnerability of water availability across India to climate change and inter-linking of rivers 2017
R. Singh, S. Veena, *Ankit Deshmukh*
(Sustainable Holistic Water Resources Management, Raju, S., and Vasan, A. (eds)., M/S Jain Brothers, New Delhi, ISBN: 978-81-8360-253-2)

Implications of river interlinking on vulnerability of water availability across India 2017
Ankit Deshmukh, R. Singh, S. Veena
(Sustainable Holistic Water Resources Management, Raju, S., and Vasan, A. (eds)., M/S Jain Brothers, New Delhi)

Patents

1. **Continuous External Vehicle Monitoring and Alerting/Warning System for Traffic Management —** Indian Patent Application No. 202421090466 A, published May 30, 2025. Developed AI-integrated system leveraging camera networks, sensors, edge computing, and machine learning for real-time traffic violation detection and safety alerts.
2. **Advanced GIS-Based Drought Prediction and Management Framework for Sustainable Agriculture in India —** Indian Patent Application No. 202521039128 A, published October 3, 2025. Engineered GIS, IoT, and AI-driven system integrating satellite imagery, sensors, and machine learning for real-time drought risk mapping and farmer advisories.

Projects

Building Resilient Communities Through Integrated Early Warning Systems for Urban Floods and Agricultural Droughts in India 2025
Funded by Indian Council of Social Science Research (ICSSR)

- **Duration:** 5 years
- **Total grant:** ₹99,00,000
- **Budget includes:** Research staff, fieldwork, equipment and study material, contingency, and workshop, seminar, and publication costs
- **Study area:** Gujarat, with comparative benchmarking across other disaster-prone states in India
- **Themes:** Early warning systems, floods, droughts, agricultural drought, GIS, remote sensing, IoT, and machine learning

Development of a Machine Learning-YOLO Framework for Automated Pavement Distress Identification Using High-Resolution Drone Imagery 2024
PDPU Student Research Project (SRP)

- **Grant:** ₹50,000

Teaching

Pandit Deendayal Petroleum University, Teaching Faculty – Gujarat, India Jan 2021 – present

- **Undergraduates:** Advanced surveying and Geomatics (18CE305T), Basic Surveying (19CE205T), Guided undergraduate project on 'Introduction of Hydrologic modeling'
- **Graduates:** Taught 'Introduction of GIS techniques and QGIS.'

Indian Institute of Technology Hyderabad, Teaching Assistant – Hyderabad, India July 2014 – Jan 2021

- Water resources engineering (CE4502)

Honors

- Recipient of research travel grant from Ministry of Human Resource Development, Govt. of India for presenting research work at the American Geophysical Union Fall Meeting 2017, New Orleans, Louisiana, USA (December 2017).
- Recipient of national fellowship from Ministry of Human Resource Development, Govt. of India to pursue doctoral studies at Indian Institute of Technology Hyderabad, India (2016-20).
- Recipient of research excellence award for research performance at IIT Hyderabad for the years 2016 and 2017.
- Recipient of Chief Minister's award for best performance in academics.

Certifications

AI For Everyone June 2026
Issuer: DeepLearning.AI (Coursera)
- Completed a 6-hour professional course covering AI fundamentals, machine learning concepts, responsible AI, and real-world applications.
- Achieved a final grade of 95%. Course instructed by Andrew Ng.
- Credential ID: HZ5UQDVMM1WW

Remote Sensing and GIS Sept 2025
issuer: NPTEL (IIT Guwahati, SWAYAM)
- Successfully completed an 8-week NPTEL certification course covering.
- Advanced remote sensing principles, GIS fundamentals, spatial analysis,geospatial applications.
- Achieved Elite certification with a consolidated score of 69%.
- Credential ID: NPTEL25CE108S635400089

Programming for Geospatial Hydrological Applications Apr 2021
Issuer: United Nations Educational, Scientific and Cultural Organization (UNESCO) – Intergovernmental Hydrological Programme (IHP)
- Completed a professional course on geospatial programming for hydrological applications.
- Covered GIS, remote sensing, spatial data management, and hydrological data analysis workflows.
- Developed skills in geospatial data processing, visualization, and computational hydrology.
- Course designed and instructed by Dr. Hans van der Kwast, IHE Delft Institute for Water Education.

Skills

Tools: Data analysis, GIS automation and scripting; R, GRASS, and QGIS

Programming: R, MATLAB, Python

Web and illustration: Interactive web apps with R-Shiny, Inkscape & Adobe illustrator, complex data visualization

Content organization: LaTeX, Markdown, Zotero

Languages: English, Hindi

Extracurricular

- Core member of student organizing committee at 'International Conference on Modeling Tools for Sustainable Water Resources Management' organized by Department of Civil Engineering, Indian Institute of Technology Hyderabad (28-29 December 2014).
- Taught data science topics, 'Basics of Python', 'Data analysis with R', 'Basics of LaTeX' to student group at IIT Hyderabad.